

NPTEL Course on "Computer System Optimization for ML"

Lecture #16

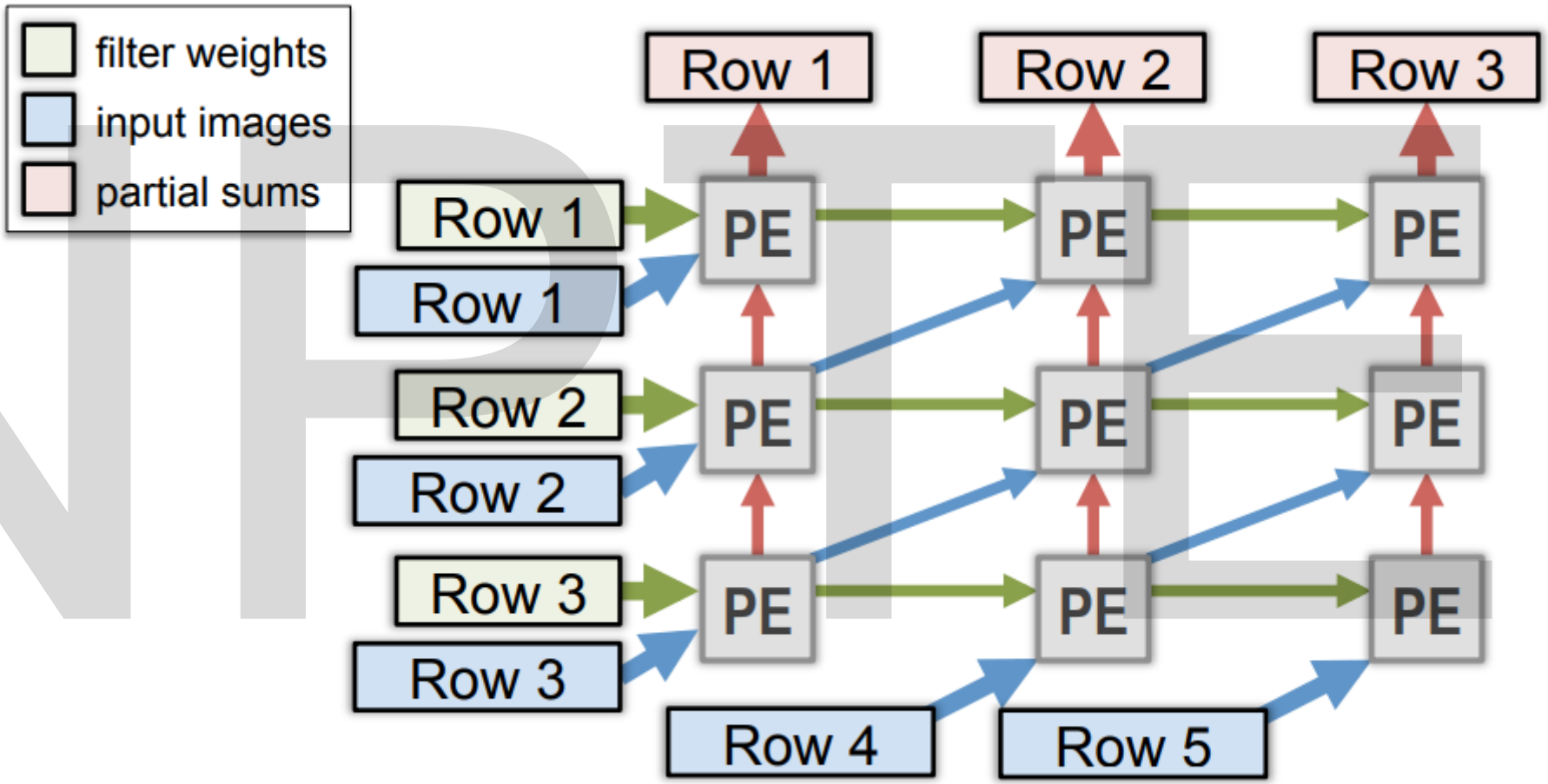
Agenda

- **Recap**
- **Eyeriss (contd)**
- **Eyeriss-V2 [1]**

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[1] Chen, Yu-Hsin, et al. "Eyeriss v2: A flexible accelerator for emerging deep neural networks on mobile devices." IEEE Journal on Emerging and Selected Topics in Circuits and Systems 9.2 (2019): 292-308.

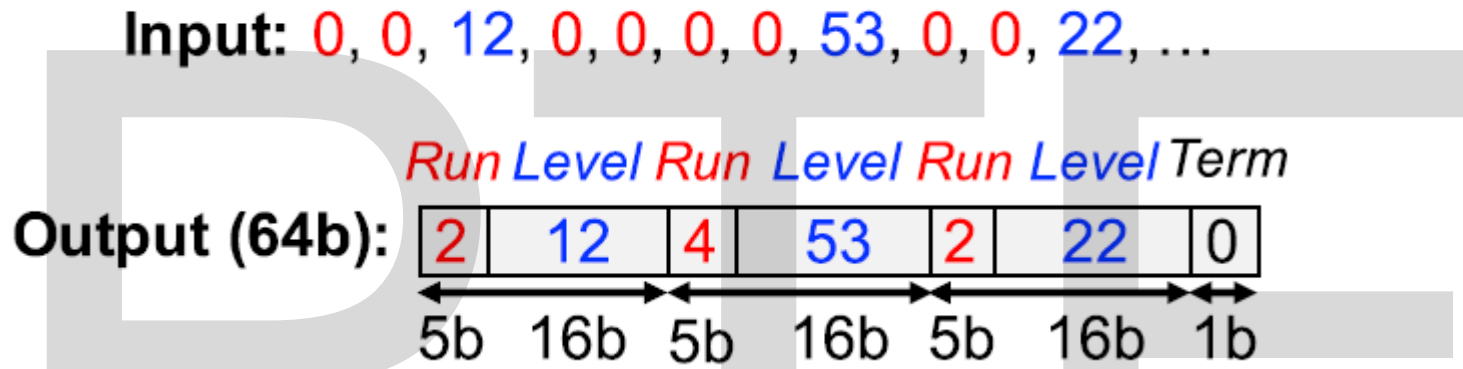
Row Stationary Dataflow (3)

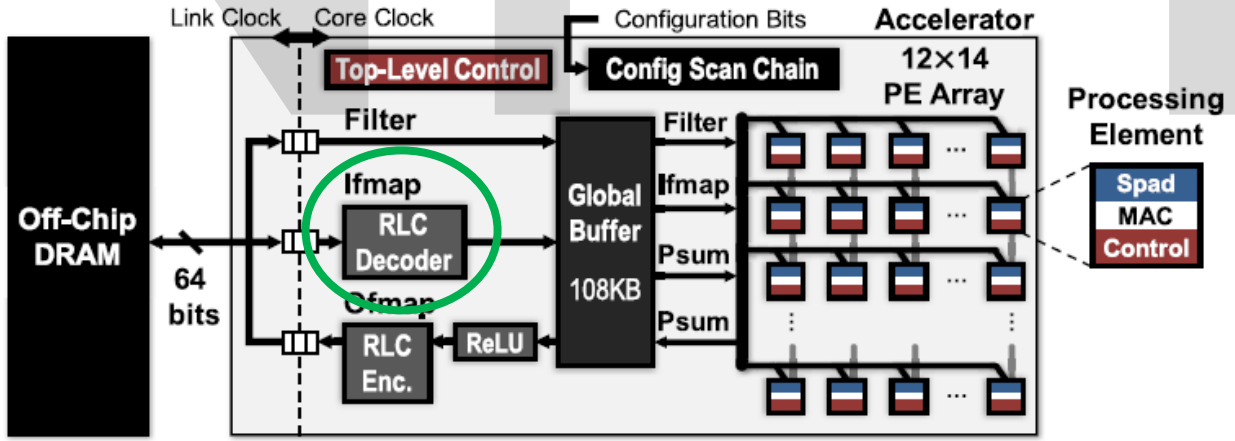


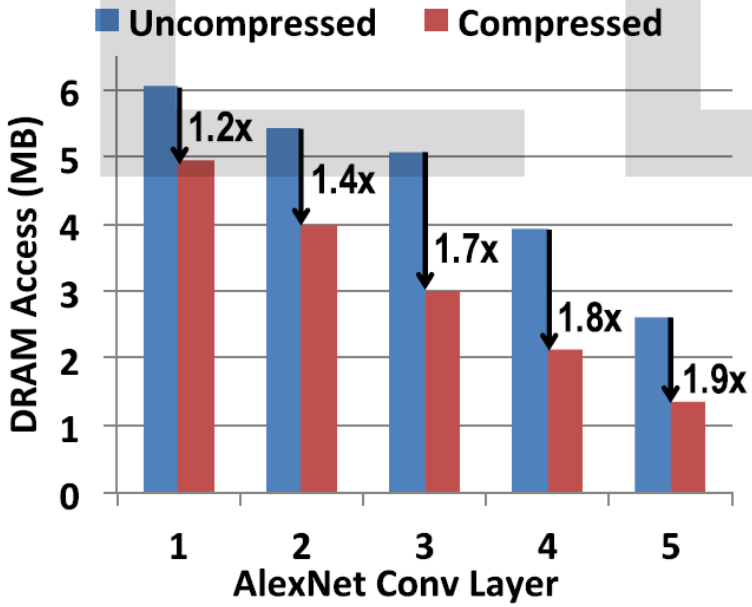
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Exploit Data Statistics: Run Level Encoding

- How should we minimize the dataflow?

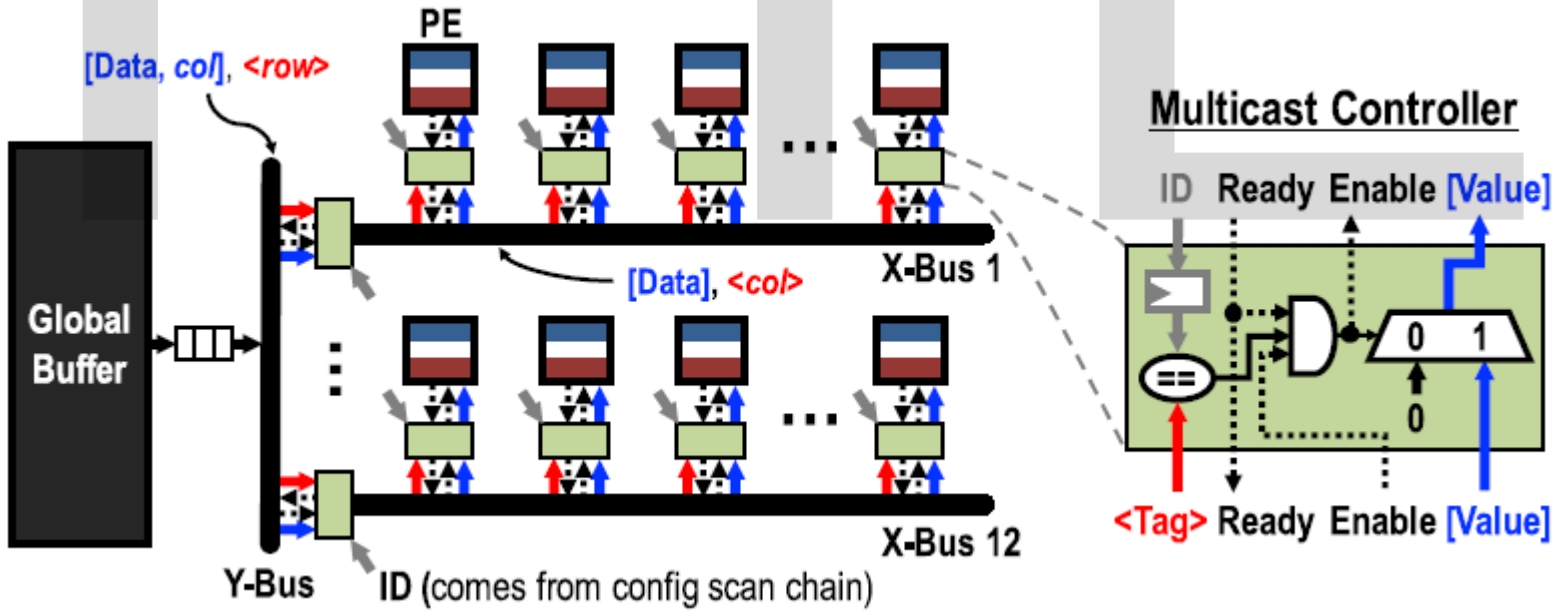






Global Input Network

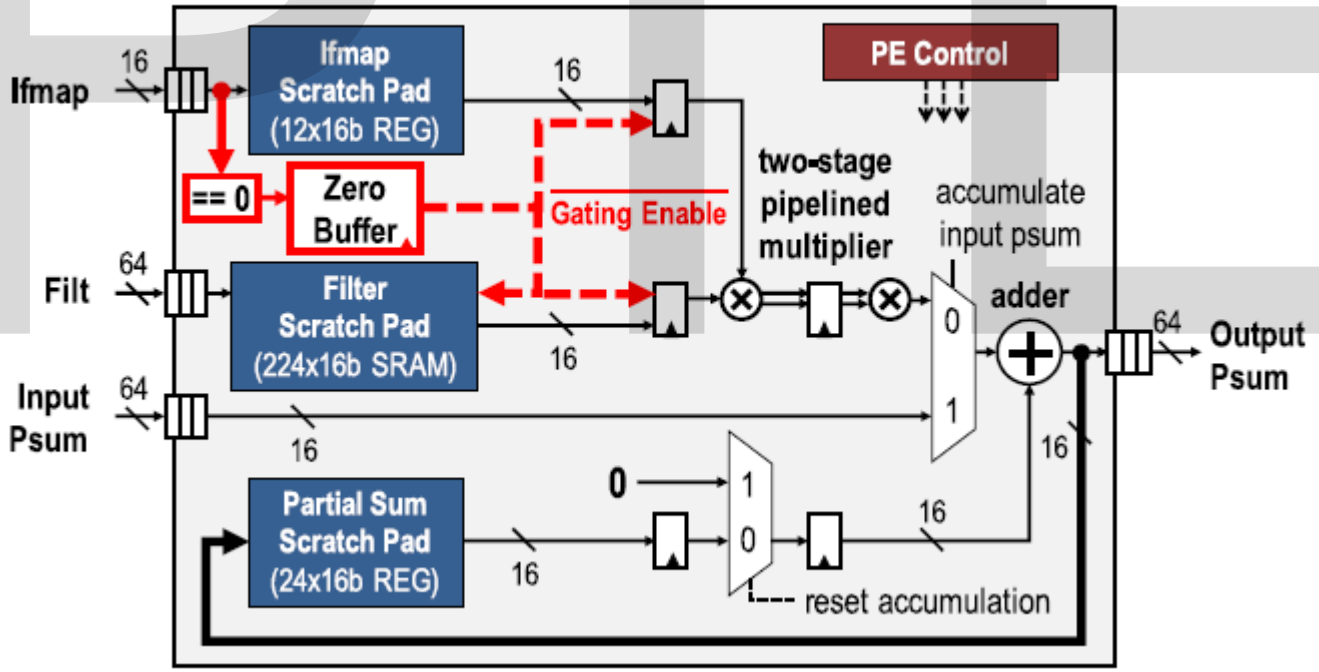
- **Consists of Y-bus and X-bus**
  - A vertical Y-bus consists of 12 horizontal X-buses
  - Each X-bus is connected to 14 PEs in each row
- **A multicast controller decodes the destination**



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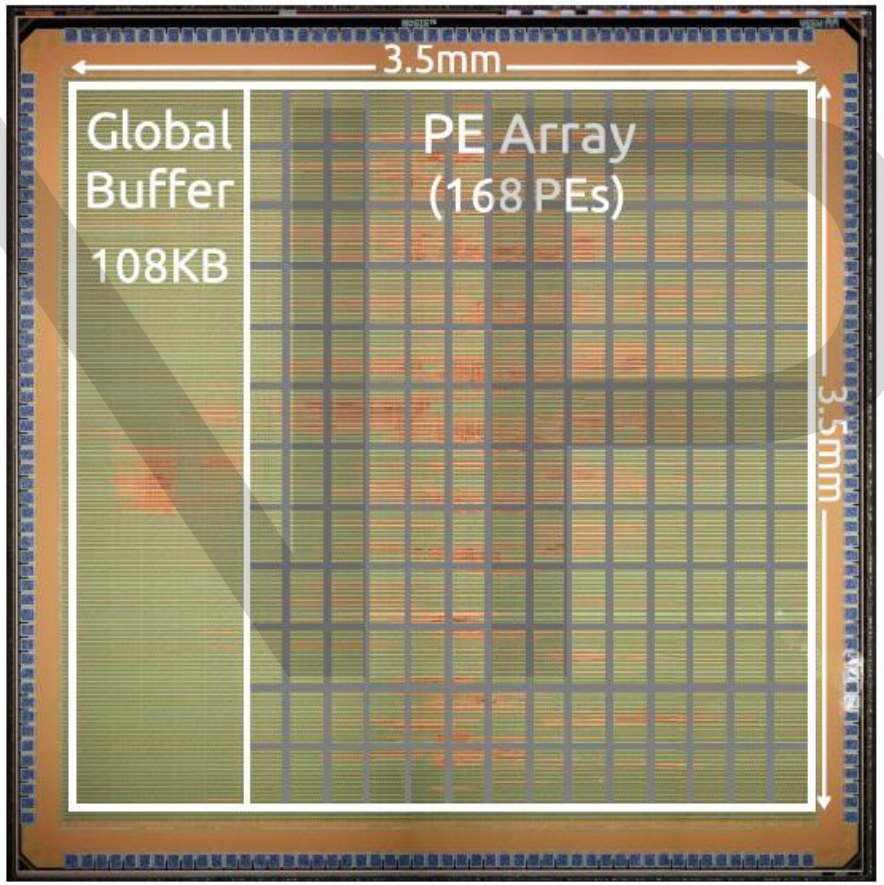
Processing Element

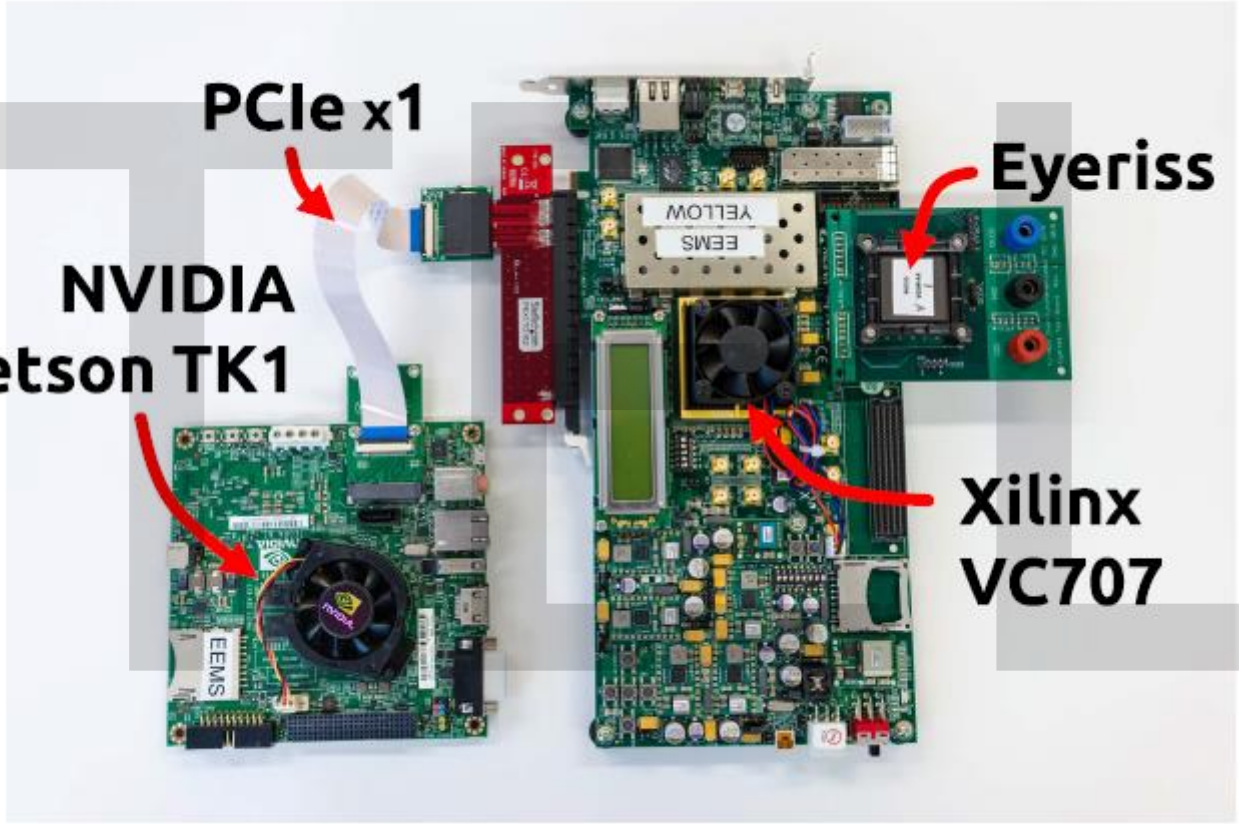
- **16-bit two-stage multiplier and adder**
- **Data gating to reduce energy consumption**
  - Shown in red dashed line



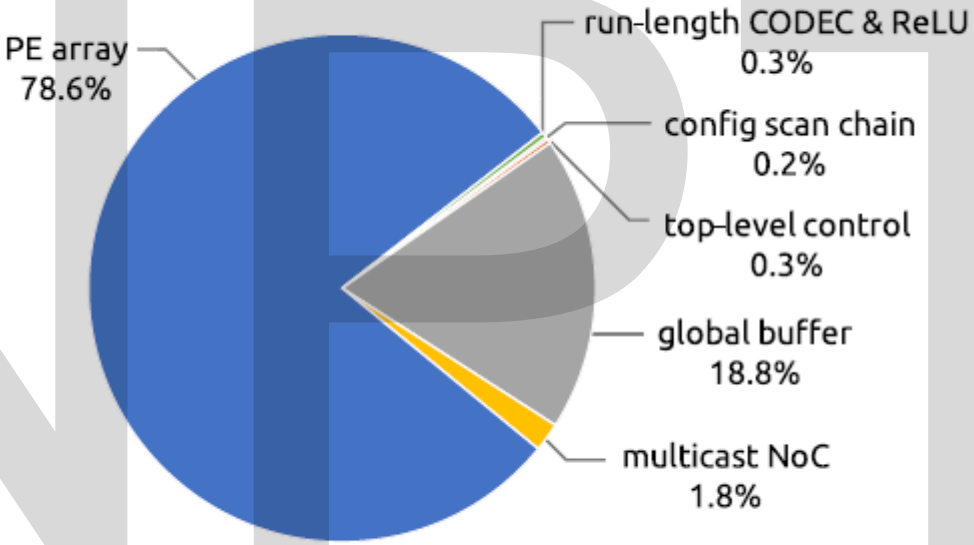
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It’s a co-processor

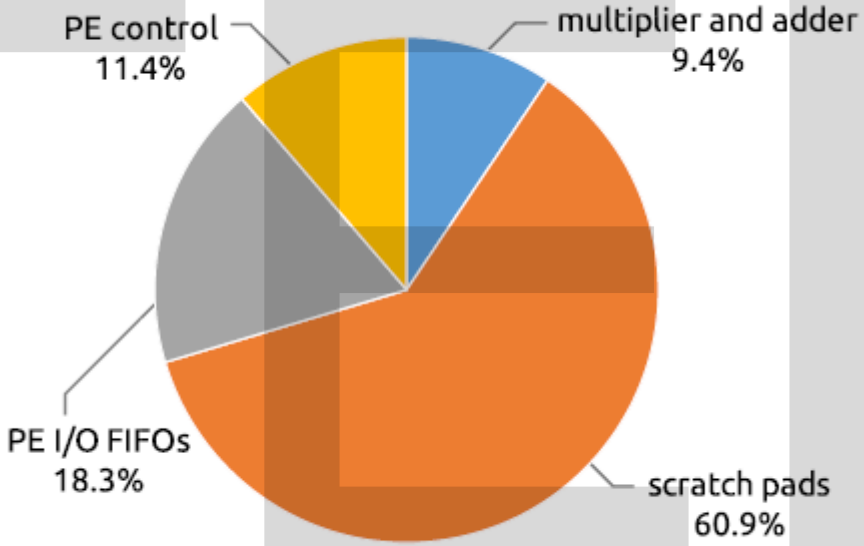




Results: Area Breakdown



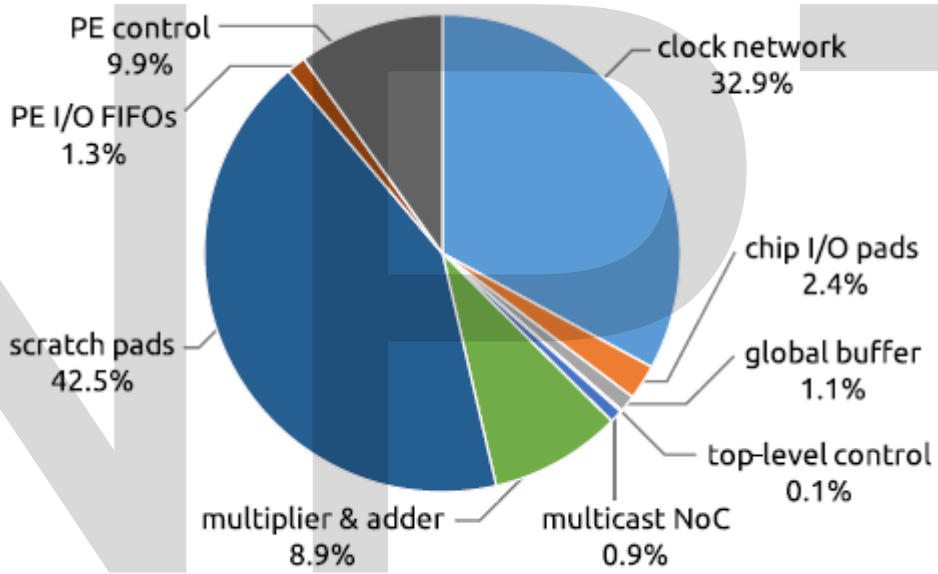
(a) **Eyeriss core**



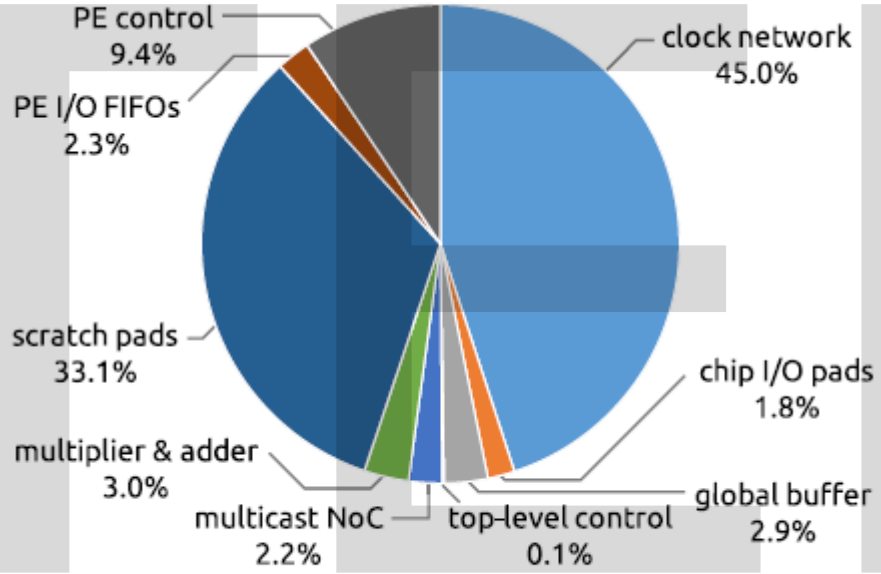
(b) **PE**

NDL

Results: Power Breakdown for AlexNet



(a) **CONV1**



(b) **CONV5**

NTL

Agenda

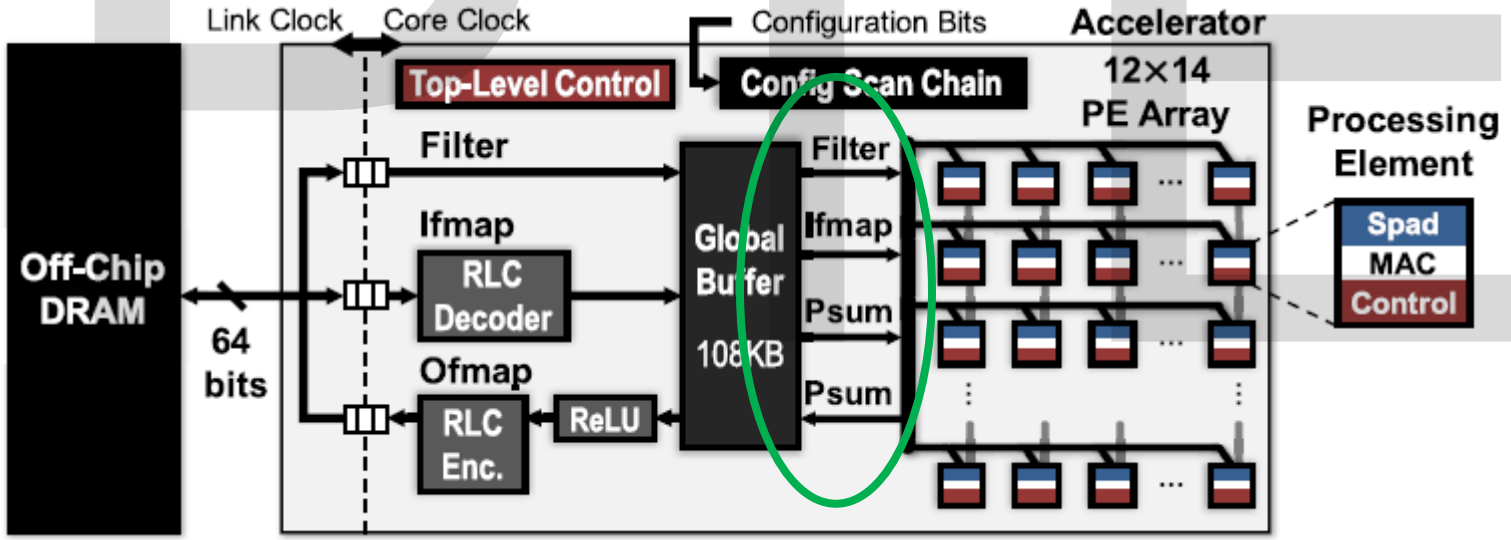
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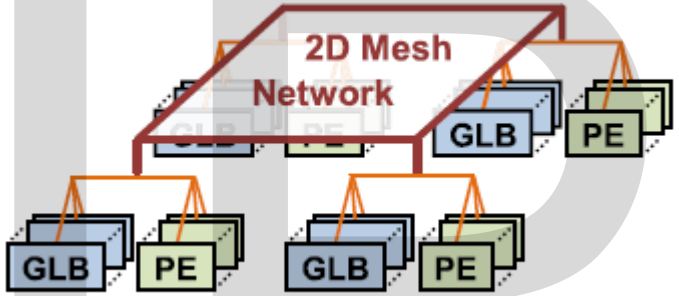
Eyeriss-V2

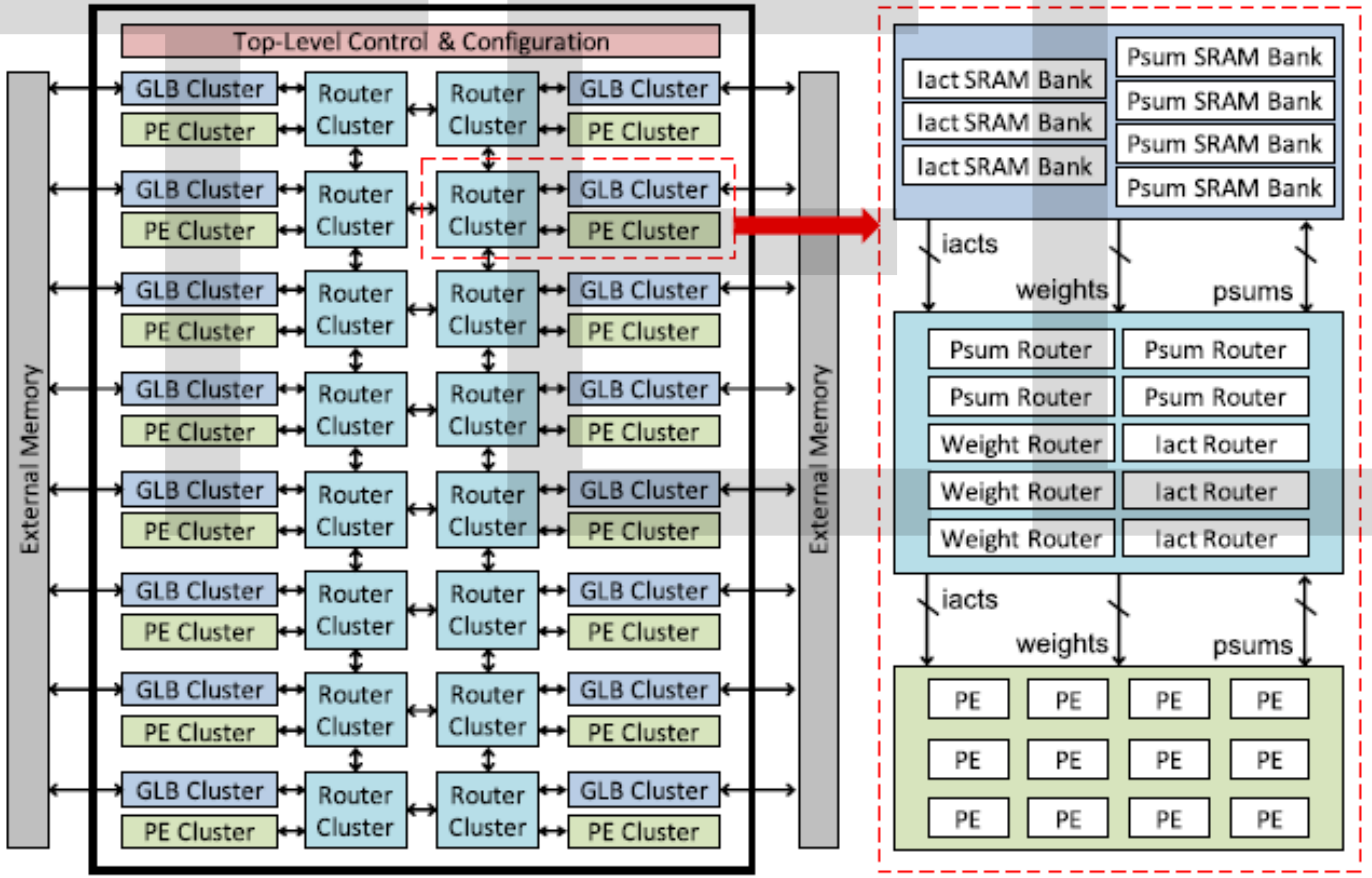
- **Eyeriss-V2: A Flexible Accelerator for Emerging DNNs on Mobile Devices**
- **What is wrong with Eyeriss?**
  - Think from communication perspective
  - Too many data transfer between global buffer and PE array



Any Idea?

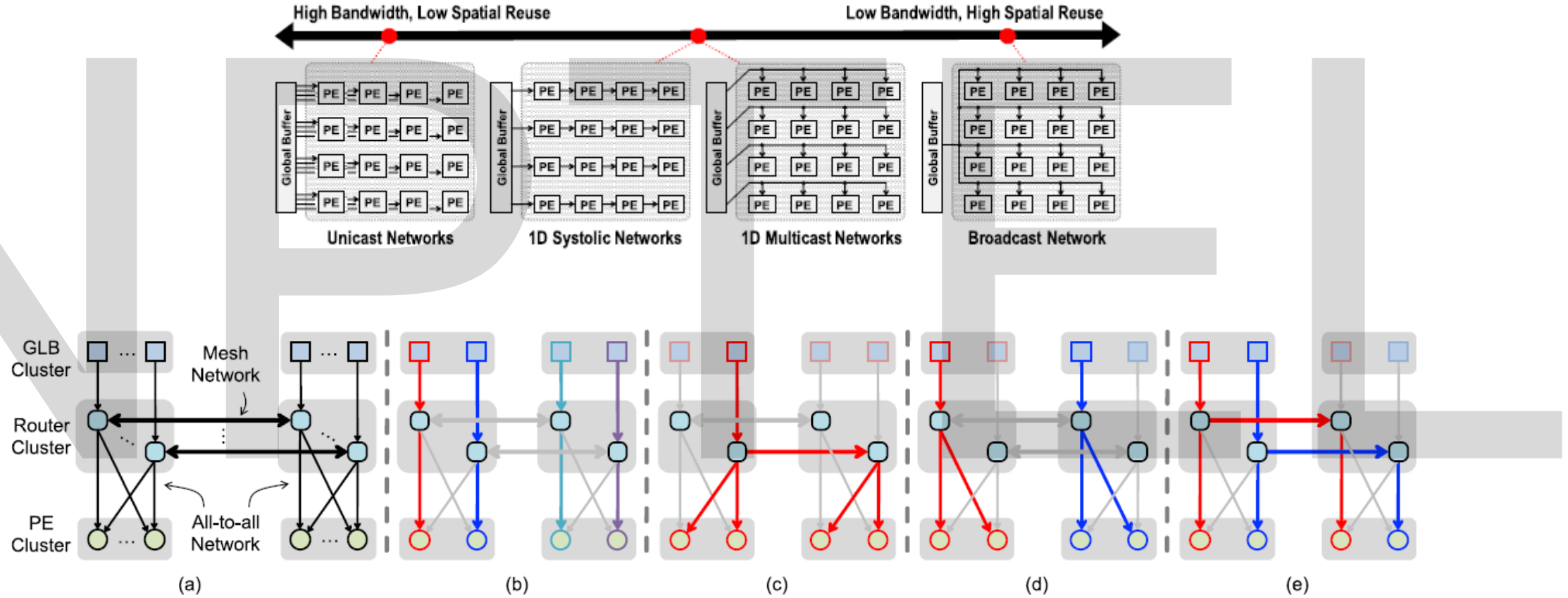
- **Distribute the communication**





Advantage

Common NoC Designs (until the paper was published)



Eyeriss-V2: a) High-level structure of the hierarchical mesh network (HM-NoC), and its different operating modes: (b) High bandwidth mode, (c) High reuse mode, (d) grouped-multicast mode, and (e) interleaved-multicast mode. In each mode, the colored arrows show the routing path; different colors denote the path for unique data.



THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)